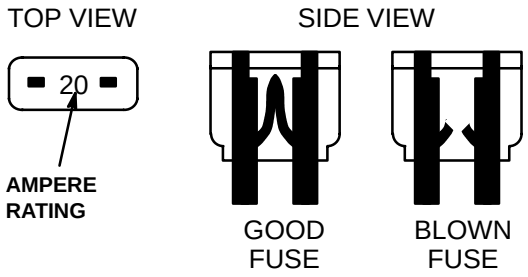
11-1 FUSE PANEL/CIRCUIT PROTECTION

CIRCUIT PROTECTION DEVICES

Electrical circuits on this vehicle may be protected by fuses, fusible links, fusible link cartridges, circuit breakers, or a combination of these devices.

BLADE TYPE FUSE

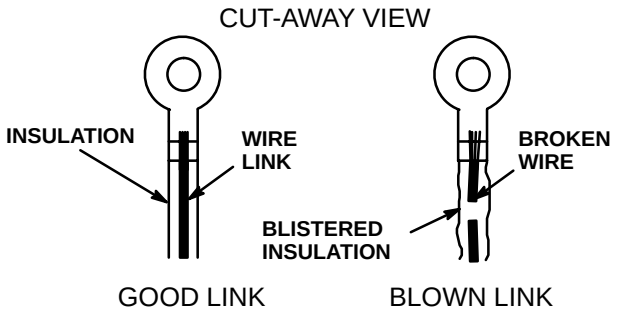


Blade type fuses have a transparent plastic housing. To check a fuse, pull it from the fuse panel and look at the fuse element through the housing. Always replace a blown fuse with a new fuse that has the same ampere rating.

The ampere rating of a blade type fuse can also be determined by following the color code shown here:

BLADE FUSE COLOR CODING	
AMPERE RATING	**HOUSING COLOR**
4	Pink
5	Tan
10	Red
15	Light Blue
20	Yellow
25	Natural
30	Light Green

FUSIBLE LINK



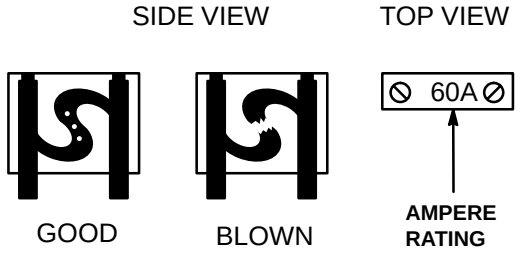
Fusible links are short lengths of wire that are smaller in diameter than the wires they are protecting. Fusible link wire is covered with a special thick, non-flammable insulation. An overload condition causes the insulation to blister. If the overload condition continues, the wire link will melt. To check a fusible link, look for blistered insulation. If the insulation is okay, pull lightly on the wire; if the fusible link stretches, the wire has melted.

When replacing fusible links, first cut the protected wire where it is connected to the fusible link. Then, tightly crimp or solder the new link to the protected wire.

Fusible links are often identified by color coding of the insulation, as shown here:

FUSIBLE LINK COLOR CODING	
WIRE LINK SIZE	**INSULATION COLOR**
20 GA	Blue
18 GA	Brown or Red
16 GA	Black or Orange
14 GA	Green
12 GA	Gray

MAXI-FUSE CARTRIDGE



Maxi-fuse cartridges have a transparent colored plastic housing. To check a maxi-fuse cartridge, look at the fuse element through the side of the housing.

To replace a maxi-fuse cartridge, pull it from the fuse box or panel. Always replace a blown maxi-fuse cartridge with a new one having the same ampere rating.

The ampere rating of a maxi-fuse cartridge can also be determined by following the color code shown here:

MAXI-FUSE CARTRIDGE COLOR CODING	
AMPERE RATING	**HOUSING COLOR**
20	Yellow
30	Light Green
40	Amber
50	Red
60	Blue

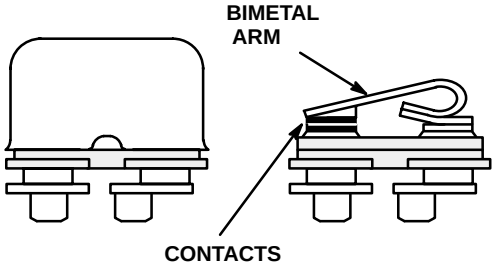
CIRCUIT BREAKER

Some circuits are protected by circuit breakers (abbreviated “c. b.” in fuse chart).
They can be Fuse Panel mounted or in-line. Like fuses, they are rated in amperes.

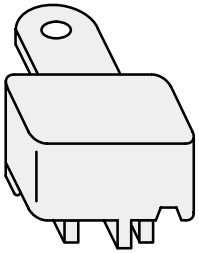
Each circuit breaker conducts current through an arm made of two types of metal bonded together (bimetal arm). If the arm starts to carry too much current, it heats up. As one metal expands faster than the other the arm bends, opening the contacts. Current flow is broken. A circuit breaker can be the cycling or non-cycling type.

FUSE PANEL MOUNTED CYCLING TYPE

SIDE VIEW (EXTERNAL) SIDE VIEW (INTERNAL)



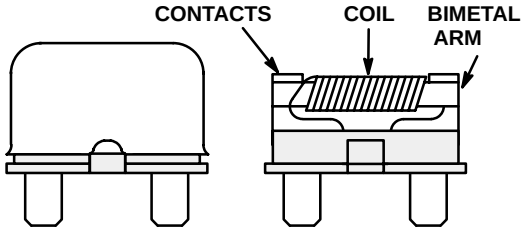
IN-LINE MOUNTED CYCLING TYPE



In the cycling type, the bimetal arm cools and straightens out. This cycle repeats as long as the overcurrent exists and power is applied.

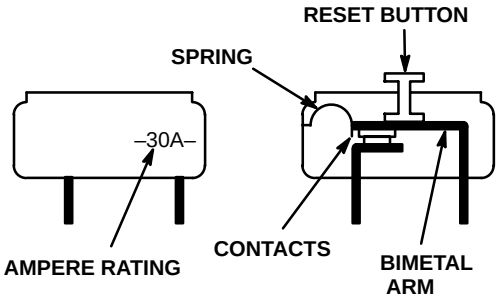
FUSE PANEL MOUNTED NON-CYCLING TYPE

SIDE VIEW (EXTERNAL) SIDE VIEW (INTERNAL)



FUSE PANEL MOUNTED MANUAL RESET TYPE

SIDE VIEW (EXTERNAL) SIDE VIEW (INTERNAL)

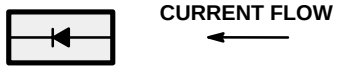


Two types of non-cycling circuit breakers are used; one is reset by removing power from the circuit, and the other is reset by depressing a reset button.

In the first type, there is a coil wrapped around the bimetal arm. When an overcurrent exists and the contacts open, a small current passes through the coil. This current through the coil is not enough to operate a load, but it does heat up both the coil and the bimetal arm. This keeps the arm in the open position until power is removed.

In the second type, a spring pushes the bimetal arm down and holds the contacts together. When an overcurrent condition exists and the bimetal arm heats up, the bimetal arm bends enough to overcome the spring and the contacts snap open. The contacts stay open until the reset button is pushed and the contacts snap together again.

DIODE



Diodes are electrical devices that permit current to flow in one direction only. The current flows in the direction indicated by the arrow.

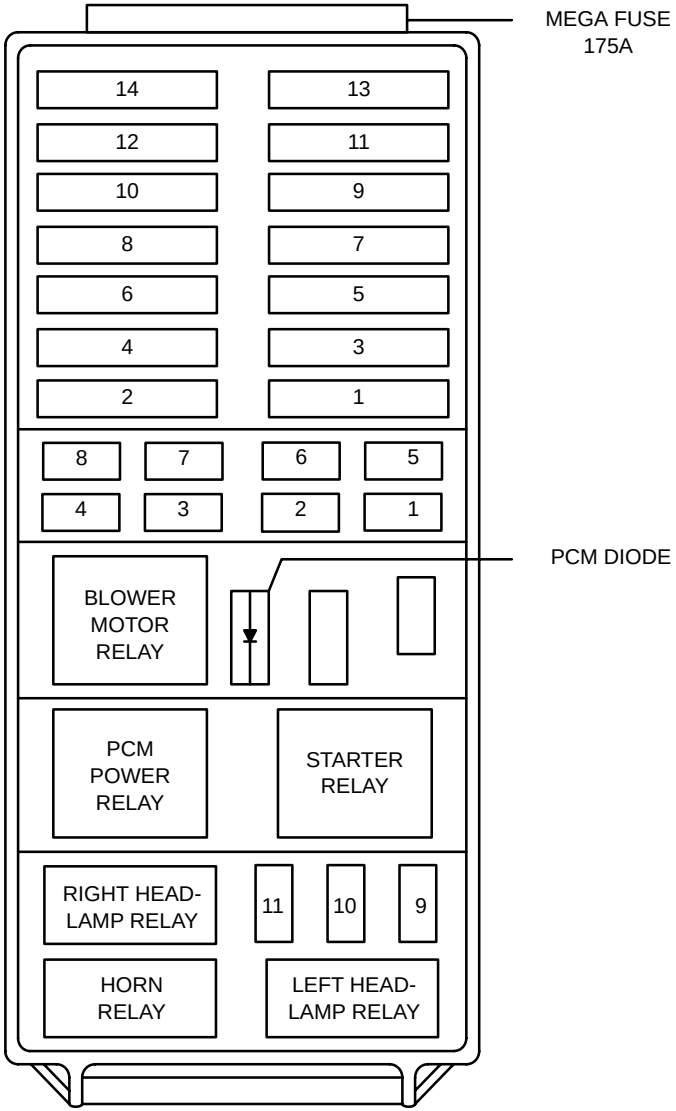
MEGA FUSE

TOP VIEW

Mega fuse protects the charging circuit. In the event that the mega fuse is blown due to the charging circuit failure, the generator field circuit is disabled.

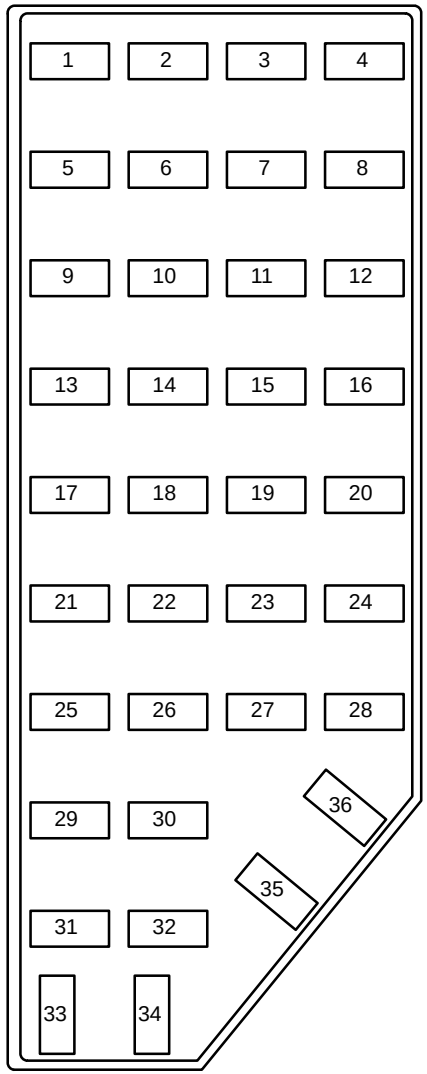
11-3 FUSE PANEL/CIRCUIT PROTECTION

BATTERY JUNCTION BOX



Maxi Fuse	Amps	Circuits Protected
1	20	4 Wheel Drive Control Module
2	30	Powertrain Control Module (PCM) Power Relay
3	20	Fuel System
4	30	Headlamps
5	50	ABS System
6	30	ABS System
7	–	NOT USED
8	–	NOT USED
9	40	Blower Motor Relay
10	–	NOT USED
11	–	NOT USED
12	–	NOT USED
13	60	Central Junction Box
14	50	Ignition Switch
Mini Fuse	**Amps**	**Circuits Protected**
1	15	Parking Lamps
2	–	NOT USED
3	–	NOT USED
4	–	NOT USED
5	10	Powertrain Control Module (PCM) Memory
6	–	NOT USED
7	15	Horn
8	–	NOT USED
9	–	NOT USED
10	30	Generator
11	15	Hego System

CENTRAL JUNCTION BOX



Mini Fuse	Amps	Circuits Protected
1	15	4 Wheel Drive Control Module
2	7.5	Blower Motor Relay
3	–	NOT USED
4	15	Left Headlamp Relay
5	10	Data Link Connector (DLC)
6	20	Utilimaster HVAC/Defrost
7	–	NOT USED
8	15	Right Headlamp Relay
9	–	NOT USED
10	7.5	Flasher Relay, Shift Lock Actuator, Multifunction Switch, 4 Wheel Drive Control Module
11	7.5	Instrument Cluster, Fuel Tank Assembly, 4 Wheel Drive Control Module Utilimaster HVAC/Defrost
12	–	NOT USED
13	15	Brake Pedal Position (BPP) Switch
14	10	4 Wheel Anti-Lock Brake System (4WABS) Module
15	7.5	Instrument Cluster
16	30	Wiper Control Module, Wiper Motor
17	25	Cigar Lighter
18	–	NOT USED

Mini Fuse	Amps	Circuits Protected
19	25	Ignition Coil, Radio Noise Capacitor, PCM Diode
20	–	NOT USED
21	15	Flasher Relay, Utilimaster HVAC/Defrost
22	–	NOT USED
23	–	NOT USED
24	7.5	DTR Sensor
25	7.5	Instrument Cluster
26	10	Main Light Switch, Utilimaster Body
27	15	DTR Sensor, Transmission Control Switch (TCS)
28	–	NOT USED
29	–	NOT USED
30	–	NOT USED
31	–	NOT USED
32	–	NOT USED
33	5	Multifunction Switch (Power Supplied through the Multifunction Switch)
34	–	NOT USED
35	7.5	Right Front Lighting Assembly, Right Turn Repeater Relay, Utilimaster Body
36	7.5	Left Front Lighting Assembly, Left Turn Repeater Relay, Utilimaster Body